

ICAS

PAPER
B



2017
SCIENCE

DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

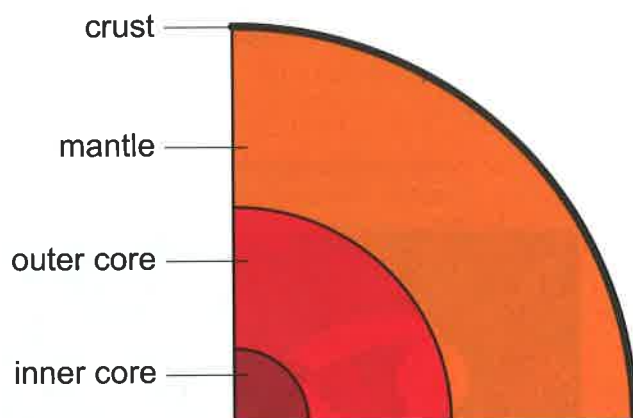
There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. The diagram shows the layers of Earth.

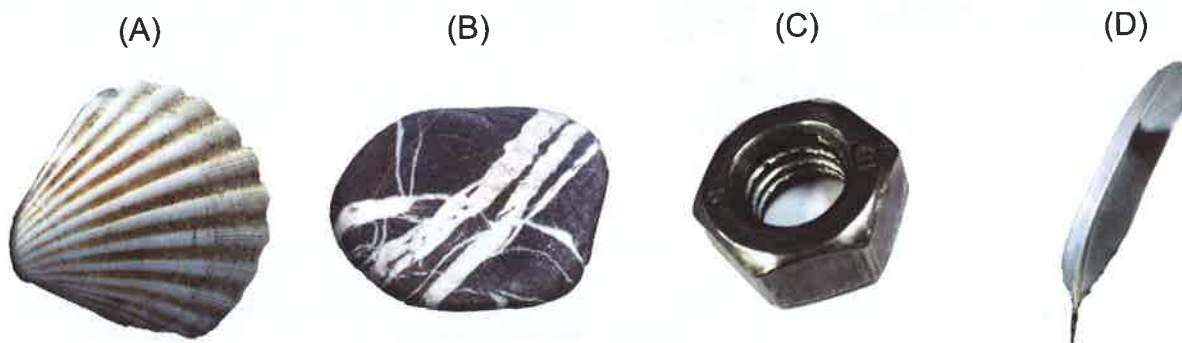


According to the diagram, which layer of Earth is thickest?

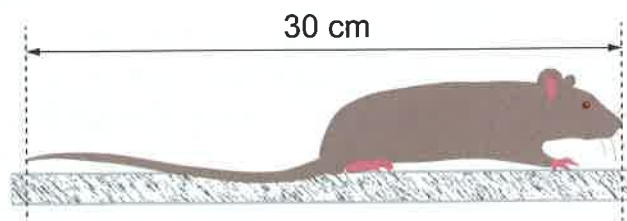
- (A) crust
- (B) mantle
- (C) outer core
- (D) inner core

2. Natural objects are things made by nature. Manufactured products are things made by humans.

Which of the following is a manufactured product?



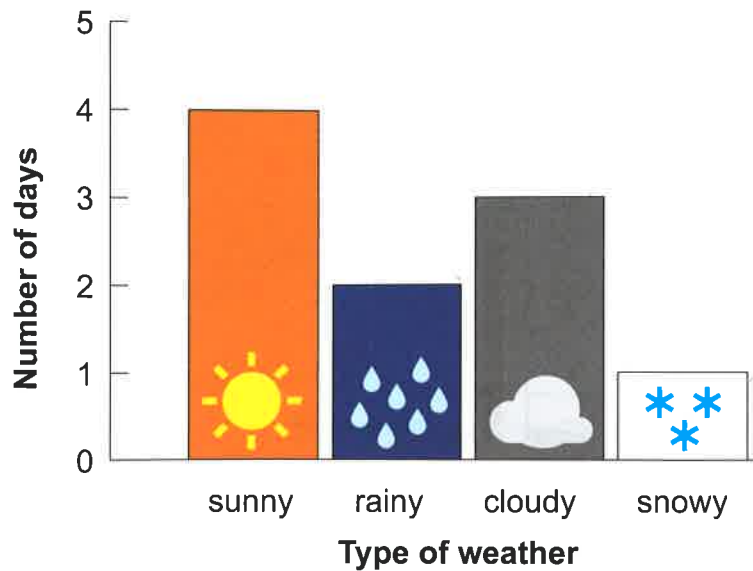
3. This is a drawing of a rat.



About how many centimetres long is the rat's tail?

- (A) 5
- (B) 10
- (C) 15
- (D) 20

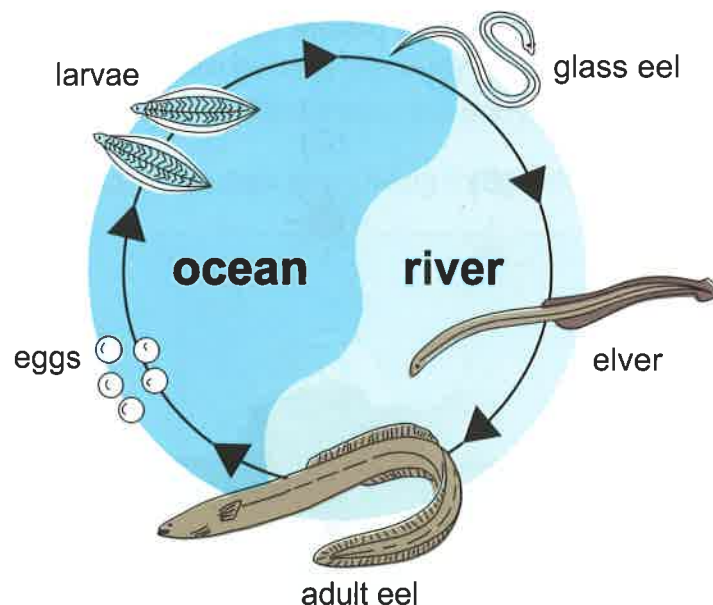
4. Ali recorded the weather on some days and used his results to draw this graph.



Over how many days did Ali record the weather?

5. An eel spends part of its life in the river and part in the ocean.

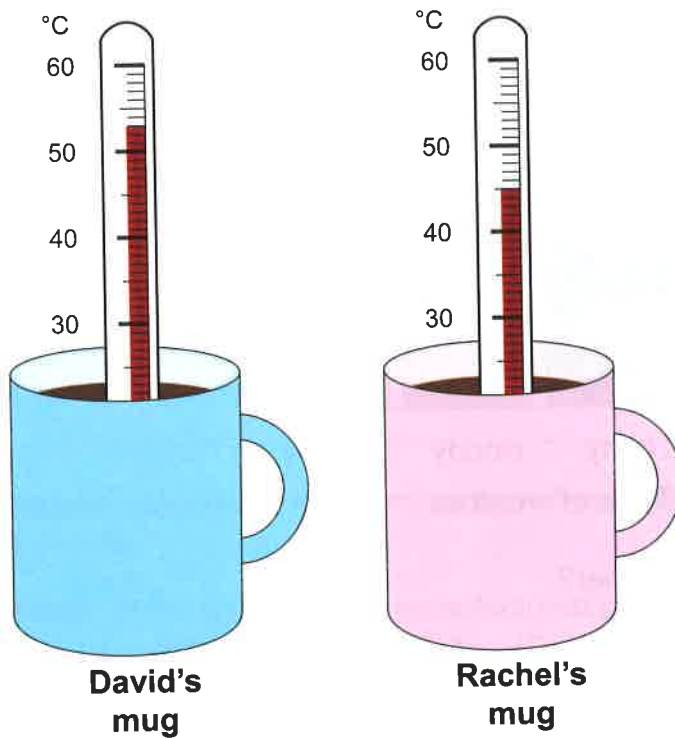
The diagram shows the life cycle of an eel.



Based on the diagram, which statement is correct?

- (A) Eggs float to the river and turn into glass eels.
- (B) Elvers live in the ocean.
- (C) Adult eels move to the ocean and lay eggs.
- (D) Larvae live in freshwater.

6. Rachel and David made hot drinks in their mugs. They placed a thermometer in each mug.

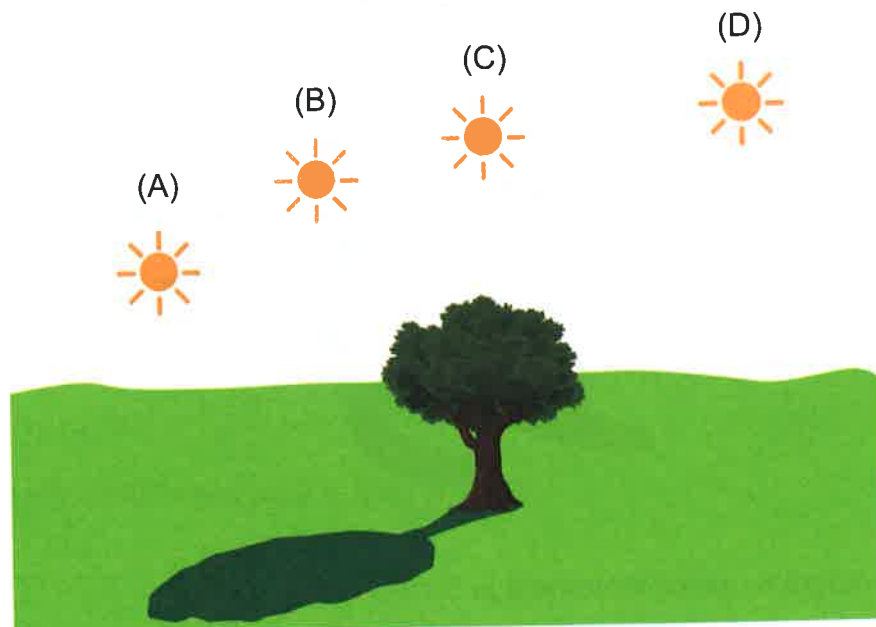


What is the difference in temperature (°C) between the two mugs of hot drink?

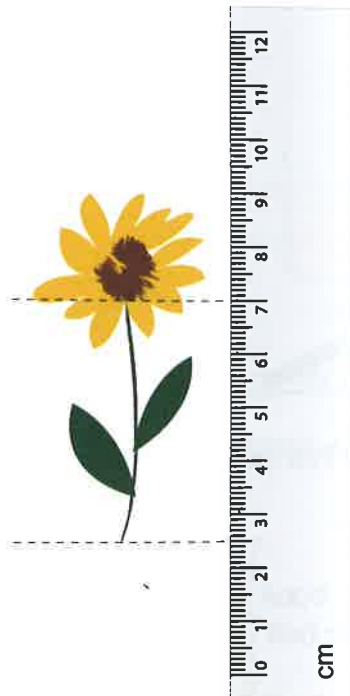
- (A) 25
- (B) 12
- (C) 10
- (D) 8

7. Alex drew a tree with its shadow.

Where must he draw the Sun to make the picture correct?



8. Kyle measured a plant using a ruler, as shown.



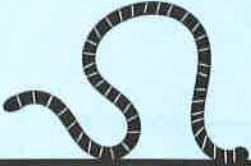
How many centimetres long is the stem of the plant?

- (A) $4\frac{1}{2}$
- (B) 5
- (C) $5\frac{1}{2}$
- (D) 7

9. Living things have different types of skeletons to support or protect their bodies. There are three main types:

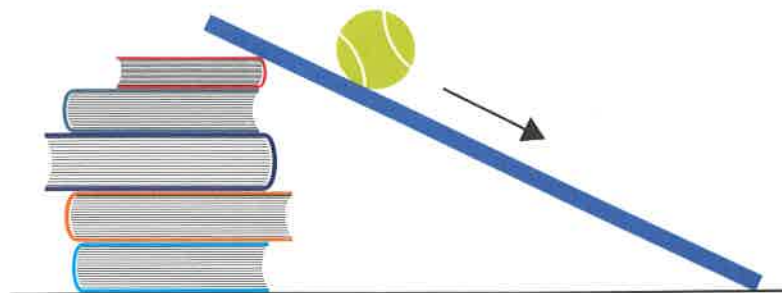
- Endoskeleton: bones inside the body
- Exoskeleton: hard structure outside the body
- Hydrostatic skeleton: tube of fluid inside the body.

Which option below correctly matches each living thing with its skeleton?

	worm 	snail 	human hand 
(A)	endoskeleton	exoskeleton	hydrostatic skeleton
(B)	hydrostatic skeleton	exoskeleton	endoskeleton
(C)	exoskeleton	hydrostatic skeleton	endoskeleton
(D)	hydrostatic skeleton	endoskeleton	exoskeleton

10. Dylan wants to investigate whether changing the slope of a ramp affects how long it takes a ball to roll down it.

He set up a ramp as shown.



Dylan can select only one extra piece of equipment to carry out this investigation.

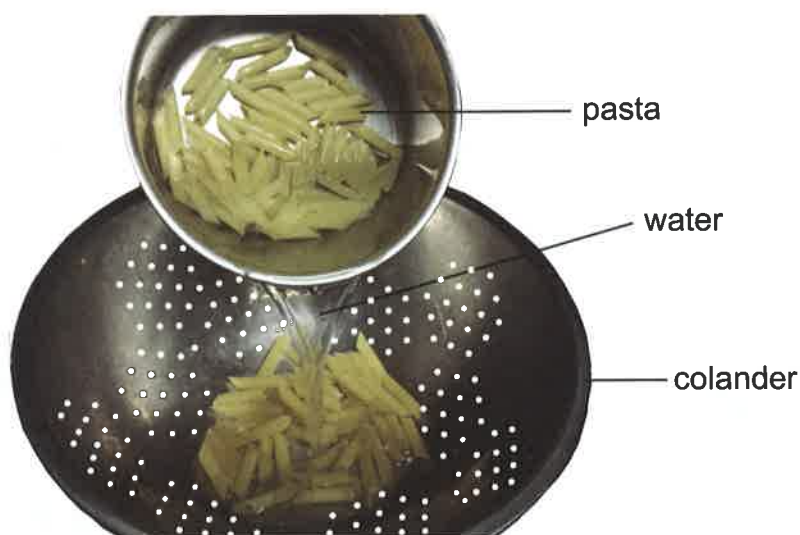
Which of the following should he select?

- (A) a stopwatch
(B) a thicker book
(C) a ruler
(D) a heavier ball

11. Filters can be used to separate solids from liquids.

The liquid that passes through the filter is called the filtrate. The solid that is left behind is called the residue.

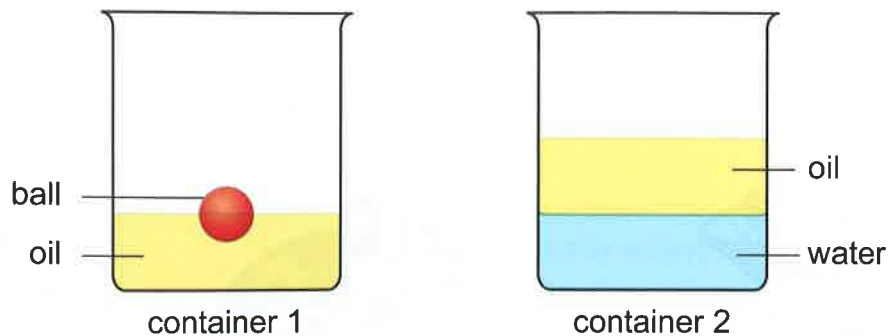
Michelle cooked some pasta in water. She then drained the water using a colander as shown.



Which option correctly matches each item with its scientific term?

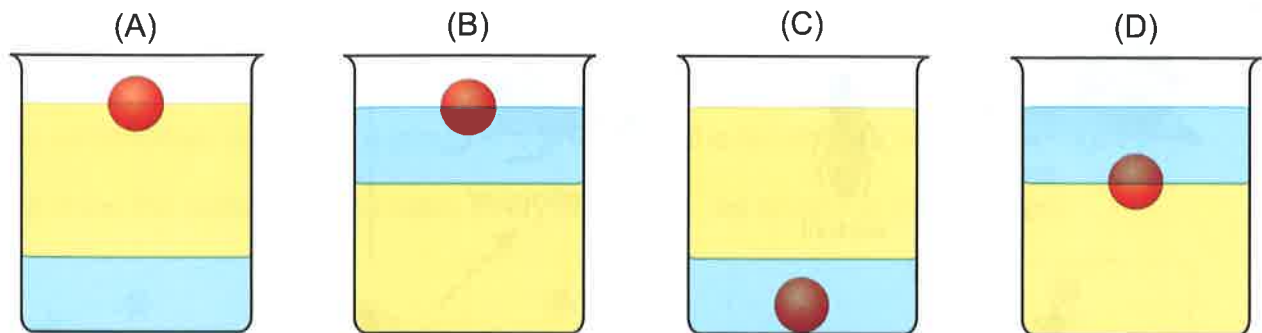
	Pasta	Hot water	Colander
(A)	filtrate	residue	filter
(B)	residue	filter	filtrate
(C)	filter	filtrate	residue
(D)	residue	filtrate	filter

12. Container 1 had oil and a small rubber ball in it. Container 2 had water and oil in it.



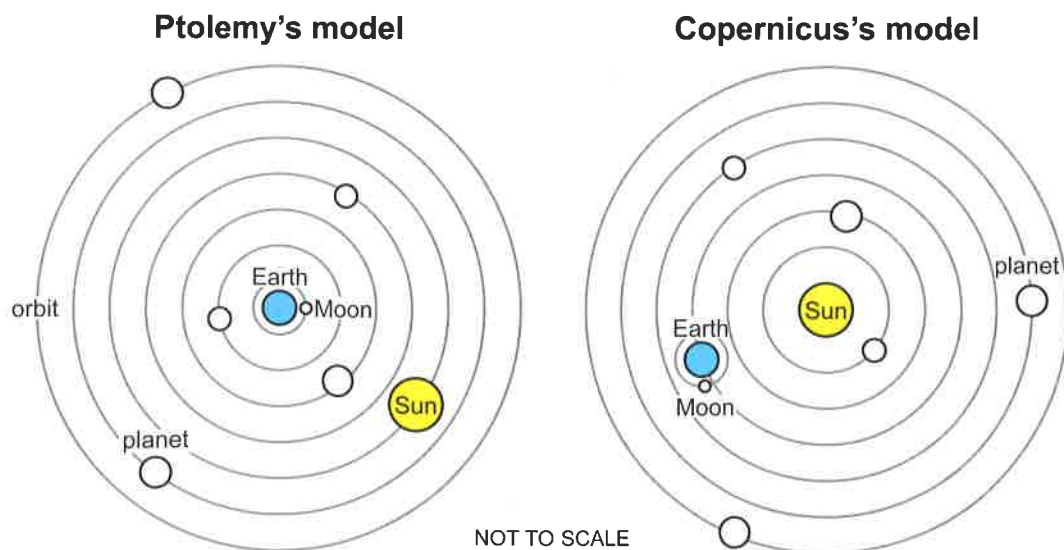
Jane poured the contents of container 2 into container 1.

Which diagram shows what container 1 looks like now?



13. Ptolemy and Copernicus were astronomers who suggested two different ways in which Earth, the Moon, the other planets and the Sun move.

An orbit represents the path taken by an object.



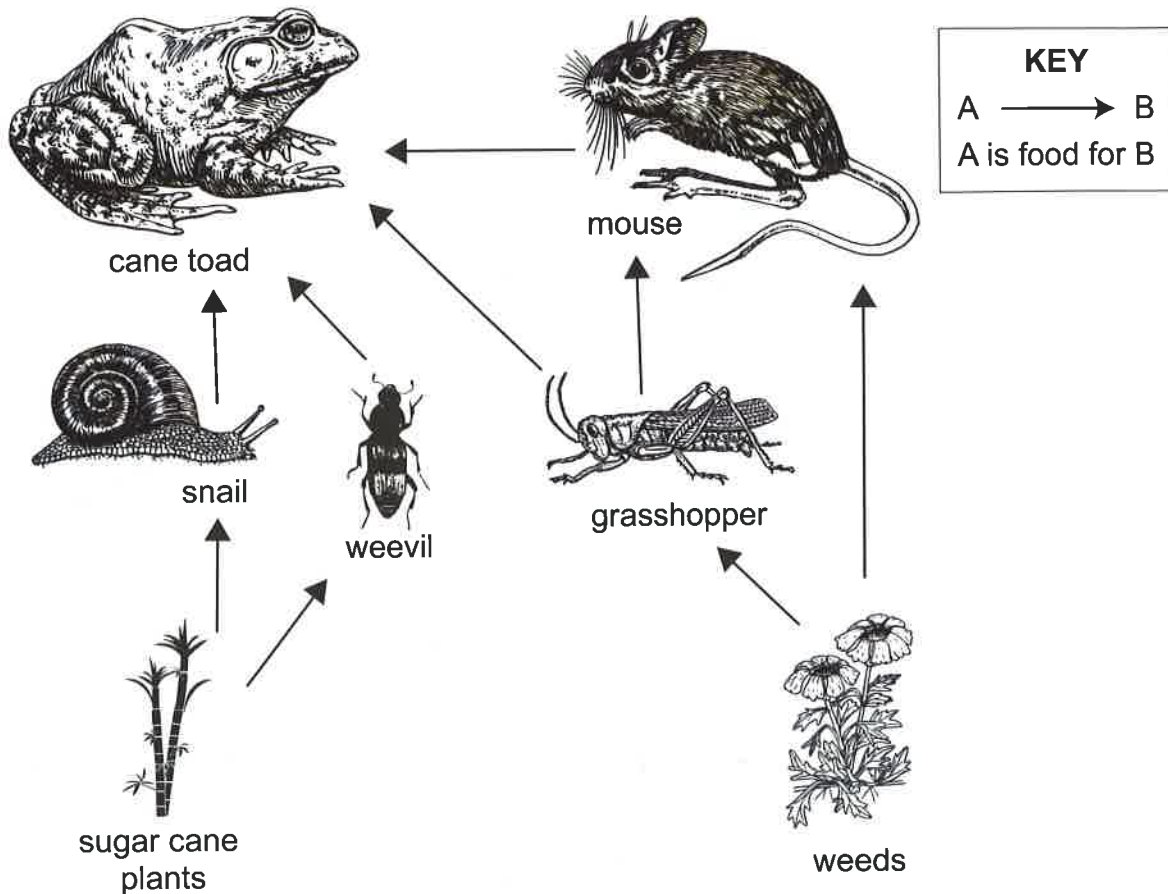
Based on the two models, which statement is correct?

- (A) Ptolemy's model suggests that the Sun is the centre of the universe.
- (B) Copernicus's model suggests that the planets orbit the Sun.
- (C) Ptolemy's model suggests that the Moon orbits the Sun.
- (D) Copernicus's model suggests that the Sun orbits Earth.

For questions 14 and 15 use the information below.

Prey is an animal that gets eaten by another animal, a predator.

The diagram represents the relationship between plants and animals living on a sugar cane farm in Australia.



14. Which of the following are predators of grasshoppers?

- (A) weeds and mouse
- (B) mouse and weevil
- (C) mouse and cane toad
- (D) cane toad and weevil

15. If all the grasshoppers were killed, which of the following is likely to happen?

- (A) The number of mice would increase.
- (B) The cane toads would begin to eat weeds.
- (C) The number of sugar cane plants would decrease.
- (D) The number of weevils and snails would decrease.

16. A key allows users to identify an object, such as an animal or a plant, using the characteristics of the object.

The diagram shows a key for a group of animals called vertebrates.

1.	has feathers	bird
	does not have feathers	go to 2
2.	has scales	go to 3
	does not have scales	go to 4
3.	has gills	fish
	does not have gills	reptile
4.	has fur	mammal
	does not have fur	amphibian

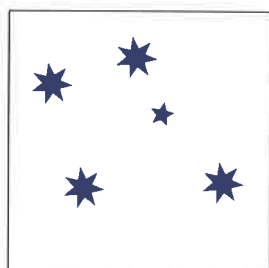
A caecilian does not have feathers, does not have scales and does not have fur.

What type of vertebrate is a caecilian?

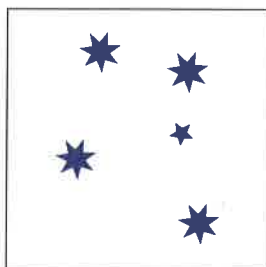
- (A) fish
- (B) reptile
- (C) mammal
- (D) amphibian

17. An astronomer observed a group of stars called the Southern Cross in the night sky.

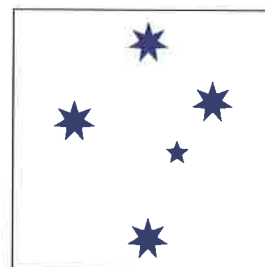
He drew the position of the stars every two hours, as shown in the diagram.



6 pm



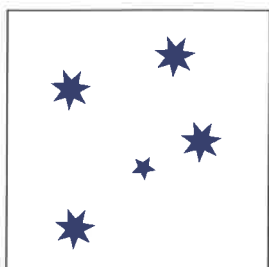
8 pm



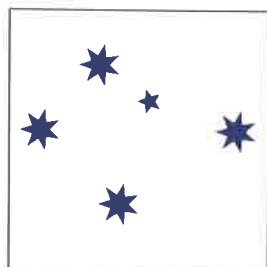
10 pm

Which of these shows the position of the Southern Cross at midnight?

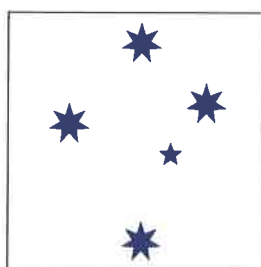
(A)



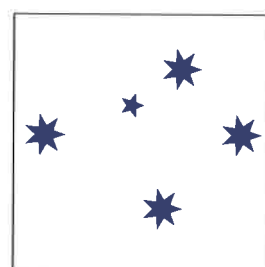
(B)



(C)

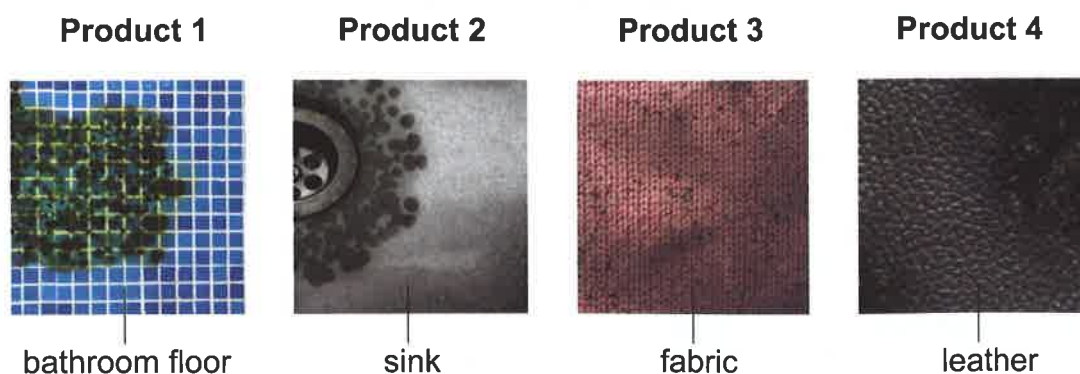


(D)



18. Angela wanted to test how well four different mould removal sprays worked.

She sprayed each product on a mouldy surface in her home.



Angela left each spray in place for ten minutes before wiping the surfaces and comparing the results.

Which of these options gives correct information about the investigation?

	Is this a fair test?	Why?
(A)	Yes	because different surfaces were tested
(B)	No	because 10 minutes was not long enough
(C)	Yes	because all methods to clean the surfaces were the same
(D)	No	because the products were not tested on the same surface

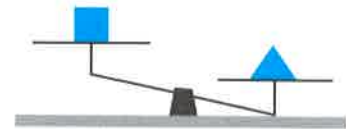
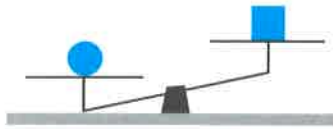
19. The table shows the dates and times for high and low tides in a harbour. These times form a pattern.

Date and time of the tide	High tide or low tide
1 July, 01:56 am	low
1 July, 08:01 am	high
1 July, 02:06 pm	low
1 July, 08:11 pm	high
2 July, 02:16 am	low

Based on the information above, what will be the date and time of the next high tide?

- | | |
|----------------------|----------------------|
| (A) 2 July, 08:21 am | (B) 2 July, 02:26 pm |
| (C) 2 July, 08:11 pm | (D) 3 July, 02:26 am |

20. Juno used a balance to weigh some blocks.



Which of the following shows the blocks in the correct order, from the lightest to the heaviest?

- (A) ● ▲ ■ (B) ■ ▲ ● (C) ■ ● ▲ (D) ▲ ■ ●

For questions 21 and 22 use the information below.

Cynthia wants to find out where at her home is quicker for drying her clothes, the leafy and windy front yard or the sunny backyard. She washes a pair of socks and at each location she hangs a sock.



front yard



back yard

21. She wants to make sure both socks are equally wet when hanging them out on the line. What is the best way to measure and compare the wetness of the socks?
- (A) weigh them
(B) compare their colours
(C) touch them to feel how wet they are
(D) squeeze them to see how much water comes out
22. Cynthia checked on the socks every 2 hours and found them both dry by the end of 4 hours. Her mum said that her results were not accurate.

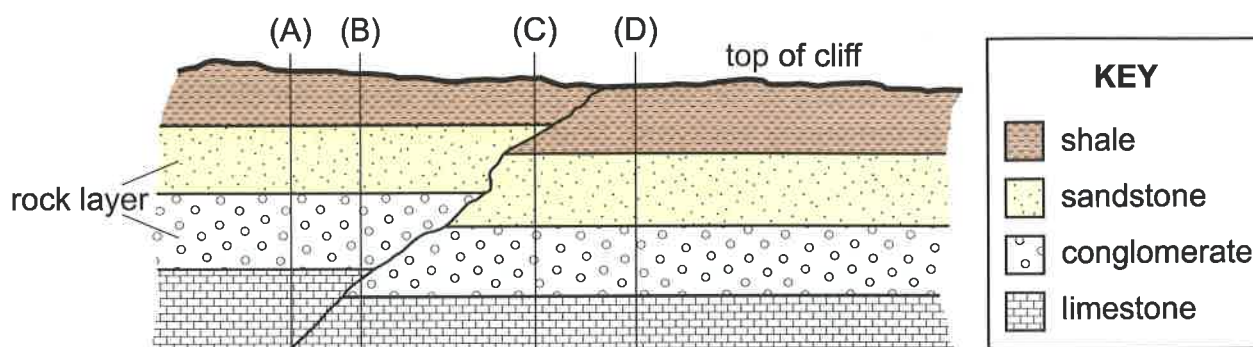
What can Cynthia do to make her results more accurate?

- (A) wash both socks using the same amount of water
(B) wash three more pairs of socks to compare
(C) check the socks every half an hour instead
(D) hang both socks at the same height

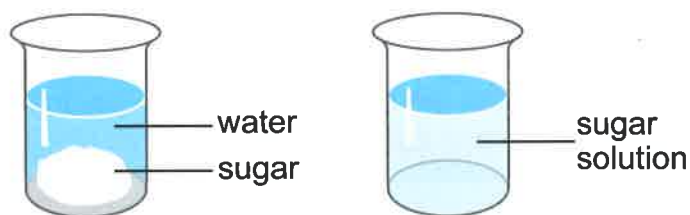
23. The diagram shows a cliff face with different layers of rock. A rock climber climbed from the top of the cliff face. She identified the following order of rocks as she moved down the cliff face:

1. shale
2. sandstone
3. conglomerate
4. limestone
5. conglomerate
6. limestone

Which path did she take down the cliff face?



24. When you add some sugar to a glass of water and stir for a while, the sugar crystals slowly disappear, leaving a clear liquid. We say the sugar dissolves in water, and the two together make a sugar solution.



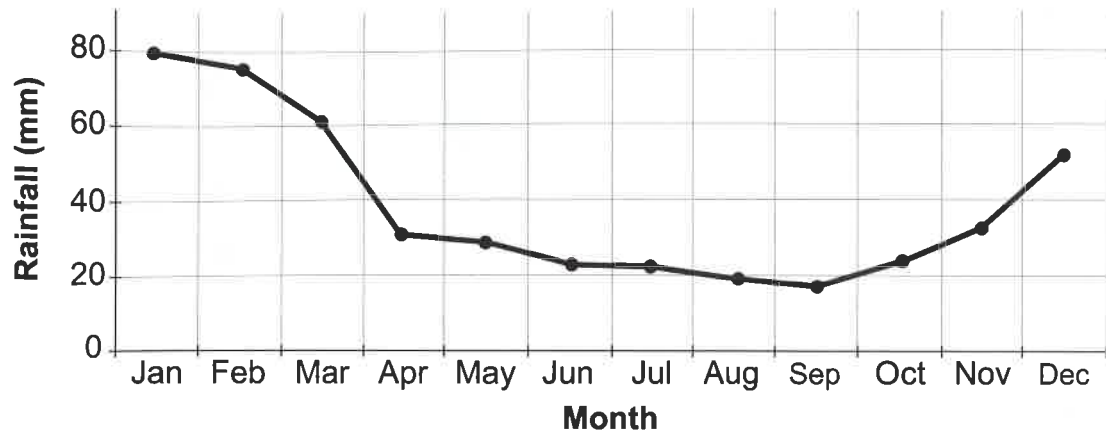
The table shows how much sugar can dissolve in 100 grams of water at different temperatures.

Temperature (°C)	10	20	40	60
Amount of sugar (grams)	180	200	240	280

How many grams of sugar can dissolve in 50 grams of water at 20 °C?

- (A) 100 (B) 150 (C) 200 (D) 250

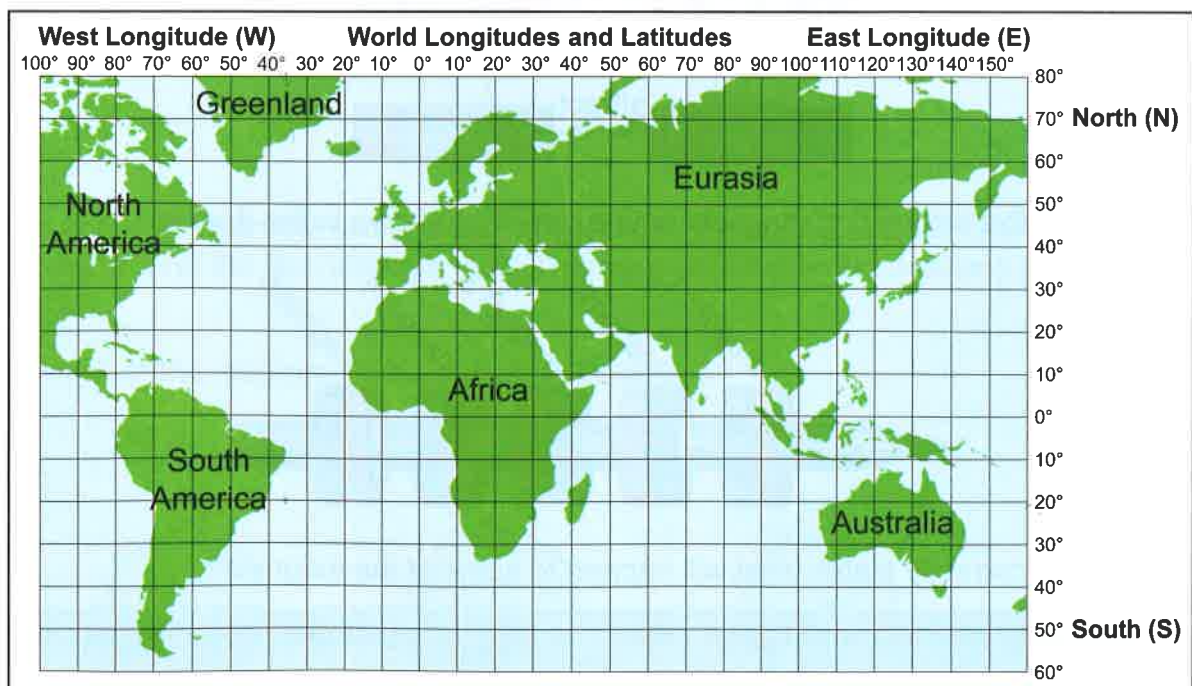
25. The graph summarises the average monthly rainfall for the years 1961–1990.



Which of the following periods had the lowest average amount of rainfall?

- (A) Jan – Mar (B) Apr – Jun (C) July – Sep (D) Oct – Dec

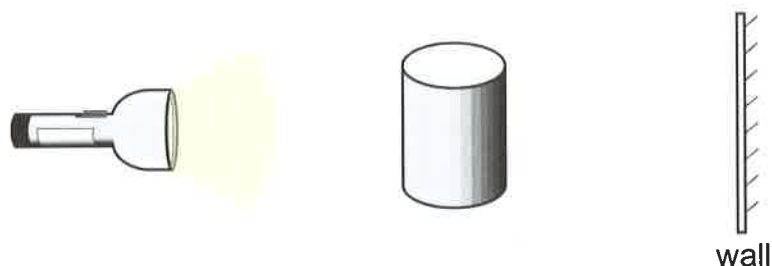
26. This is a picture of the fossil skull of *Cynognathus* that was found at location 20° S and 45° W.



Where was the fossil found?

- (A) Eurasia (B) Australia
(C) Africa (D) South America

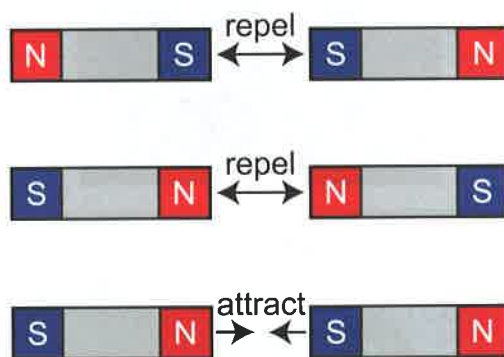
27. Joyce shines a torch onto a cylinder, as shown.



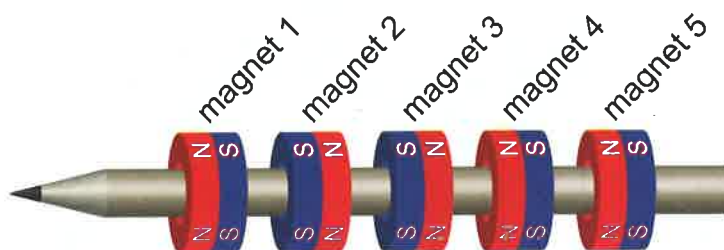
Which diagram shows the shadow produced by the cylinder on the wall?



28. Magnets have a north pole and a south pole. When magnets are placed near one another, they can either attract or repel one another as shown in the diagram.



Frans placed five strong ring magnets onto a pencil as shown in the diagram.



Which option correctly states what will happen to some of the magnets?

	Magnets that will repel each other	Magnets that will attract each other
(A)	magnet 1 and magnet 2	magnet 3 and magnet 4
(B)	magnet 3 and magnet 4	magnet 2 and magnet 3
(C)	magnet 4 and magnet 5	magnet 1 and magnet 2
(D)	magnet 2 and magnet 3	magnet 4 and magnet 5

29. Mika's class is doing an experiment using some weights and rubber bands.

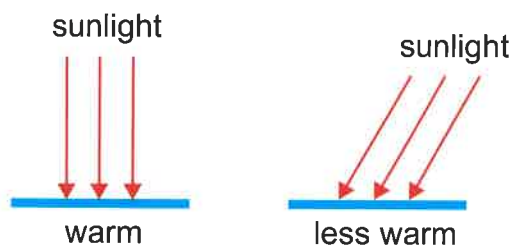
Mika has two weights and two rubber bands of the same length but of different widths. The weights can hang from the rubber bands.



Mika wants to find out whether the width of a rubber band affects how much it is stretched when a weight hangs from it.

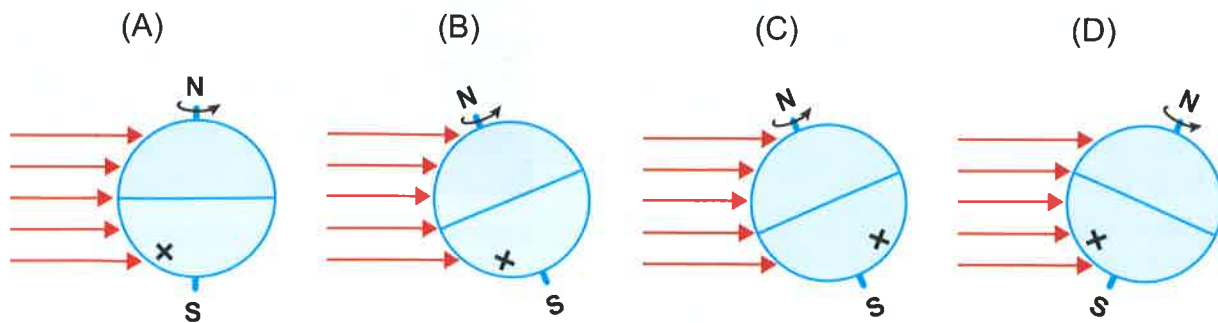
Which of the following items are required to investigate this?

- (A) both rubber bands and both weights
 - (B) both rubber bands and one weight
 - (C) one rubber band and both weights
 - (D) one rubber band and one weight
30. When a surface receives sunlight, the angle at which the sunlight strikes the surface affects the amount of heat the surface receives.



The angle at which the sunlight strikes a location on Earth's surface changes as Earth moves around the Sun during a year. This causes the different seasons at a location on Earth.

In which of the diagrams is it warmest at location x?



ACKNOWLEDGMENT

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SOURCES

Question 26 *Cynognathus* image adapted from an image by Ghedoghedo <<https://en.wikipedia.org/wiki/File:Cynognathus.JPG>> licensed under the Creative Commons Attribution-ShareAlike 3.0 Unported licence <<https://creativecommons.org/licenses/by-sa/3.0/>>.

THE FOLLOWING YEAR LEVELS SHOULD SIT THIS PAPER:

Australia	Year 4
Brunei	Primary 4
Hong Kong	Primary 4
Indian Subcontinent ¹	Class 4
Indonesia	Year 5
Malaysia	Standard 4
Middle East ²	Class 4
New Zealand/Pacific ³	Year 5
Singapore	Primary 3
Southern Africa ⁴	Grade 4

¹ Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh.

² Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Egypt, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria and Jordan.

³ Pacific Region: Vanuatu, Papua New Guinea and Fiji.

⁴ Southern African Region: South Africa, Botswana, Lesotho, Swaziland, Zimbabwe and Namibia.

PAPER

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AUSTRALIA

ANALYSIS BY QUESTION

The following table shows the questions you answered correctly ☒ and shades the questions that you answered incorrectly ☐. The questions are ranked from hardest at the top to easiest at the bottom.

	Question content	Area assessed	Question number	Correct answer
Difficult Questions	Identify a possible outcome of a disturbance in a habitat described by a food web	Predicting/Concluding	15	D
	Predict the shape of the shadow that a given object will cast	Reasoning/Problem solving	27	C
	Perform a calculation using data from a table	Interpreting	24	A
	Identify the materials required for an investigation	Investigating	29	B
	Identify the factor that improves the accuracy of an investigation	Investigating	22	C
	Determine a method to carry out a non standard measurement	Investigating	21	A
	Identify the sequence of rock strata in a cliff	Interpreting	23	B
	Rank objects on scales by mass from smallest to greatest	Reasoning/Problem solving	20	B
	Recognise the relative density of a solid and two liquids	Reasoning/Problem solving	12	A
	Determine a length using a ruler	Observing/Measuring	8	A
	Identify whether a described investigation was a fair test	Investigating	18	D
	Identify the piece of equipment needed to carry out a particular experiment	Investigating	10	A
	Identify the diagram that correctly represents a given situation	Predicting/Concluding	30	D
	Describe a household situation using scientific terminology	Investigating	11	D
	Locate a fossil using map coordinates	Observing/Measuring	26	D
Easy Questions	Make a prediction using information in diagrams	Reasoning/Problem solving	17	A
	Recognise which magnets will attract or repel	Reasoning/Problem solving	28	B
	Identify the thickest layer of Earth in a diagram	Observing/Measuring	1	B
	Use a pattern in the data to make a prediction	Predicting/Concluding	19	A
	Identify which living things have the characteristics described in presented data	Interpreting	9	B
	Estimate half of a length given a full length	Observing/Measuring	3	C
	Use a key to classify a caecilian	Predicting/Concluding	16	D
	Interpret two pictorial models of the universe	Reasoning/Problem solving	13	B
	Perform a calculation using data in a bar graph	Interpreting	4	B
	Identify a relationship in a food web	Interpreting	14	C
	Find the difference between two thermometer readings	Observing/Measuring	6	D
	Interpret a diagram showing the life cycle of an eel	Predicting/Concluding	5	C
	Determine the position of the Sun based on a shadow	Predicting/Concluding	7	D
	Identify a specified value range in a line graph	Interpreting	25	C
	Distinguish between natural materials and a synthetic material	Interpreting	2	C